

Algorithms and Datastructures — Exercise Sheet 2

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1.O-notation

- a.true
$$2n^3 + 4n^2 + 7\sqrt{n} \leq 2n^3 + 4n^3 + 7n^3 = 13n^3$$
- b.false
$$\lim_{n \rightarrow \infty} \frac{n \log_3 n}{n \log_5 n} = \frac{\ln 5}{\ln 3} < \infty.$$
$$n \log_3 n = \Theta(n \log_5 n)$$
- c.true
$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$$
- d.false
$$2 \log_2(n^2) = 4 \log_2 n$$
$$4 \log_2 n \leq c \cdot (\log_2 n)^2.$$
- e.true
 - $\max\{f(n), g(n)\} \leq f(n) + g(n)$
 - $\max\{a, b\} \geq \frac{a+b}{2}, \max\{f(n), g(n)\} \geq \frac{1}{2}(f(n) + g(n))$

2.Sorting by asymptotic growth

- $\sqrt{\log n}$
- $\log(\sqrt{n}) =_O (\log n) =_O (\log(n^3))$
- $(\log n)^2$
- $\sqrt{n}$
- $10^{100}n$
- $n \log n$
- $(2n)^2$
- n^{100}

- 2^n
- 3^n
- $n!$
- $(n + 1)!$
- n^n
- 2^{n^2}

3. Event Scheduling

Greedy algorithm

- Sort all events in non-decreasing order of their ending time t_i .
- For every next event, if i can attend(Timing doesnt conflict), then take the event.

For Greedy algorithm, the complexity is $O(n \cdot \log(n))$

- Sorting:
 $O(n \cdot \log(n))$
- Selection loop:
 $O(n)$